Design and Analysis of 4 Bit Multipliers using Pass Transistor Logic and Gate Diffusion Input Technique using 18nm FinFET Technology

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Abstract—**Very Large-Scale Integration (VLSI) is a method for manufacturing integrated circuits, where an extensive number of transistors are intricately placed onto a single silicon chip. Complementary Metal Oxide Semiconductor (CMOS) stands out as a leading technology in computer chip design, while Pass-Transistor Logic (PTL) serves as a prevalent logic design technique, utilizing transmission gates or pass transistors instead of traditional inverters for logic gate implementation. The Gate Diffusion Input Technique (GDI) is known for speed and power efficiency. A 4 x 4 array multiplier (AM) using Ripple-Carry Adder (RCA) finds applications in digital signal processing and computing, particularly in tasks like digital filtering, signal compression, image processing, and video processing. The speed of the 4 x 4 AM using RCA is slower than the other multiplication methods that incorporate Carry-Look Ahead Adder (CLA). This work involves the design of 4 x 4 AM using FinFET technology with PTL logic. The parameters like propagation delay, power, and power delay product for 4 x 4 AM is estimated using the CADENCE Virtuoso Tool at 18nm technology node with a supply voltage of 1.8V. The existing AM using RCA has the power consumption of 724.43 µW and propagation delay of 30.26 ns. The proposed AM using CLA with PTL logic experiences a delay of 10.10 ns and power consumption of 185.07 µW. This indicates that the proposed multiplier exhibits less delay and lower power consumption compared to other FinFET based multipliers. In addition, the proposed multiplier enables a maximum of 66.62 % improvement in speed and 79.12 % improvement in power when compared to its counterparts thereby showcasing its potential for improved efficiency in integrated circuit applications.**

Keywords—FinFET – PTL – GDI – Array Multiplier – Wallace tree multiplier – Power – Delay

I. INTRODUCTION

VLSI came to the forefront in the late 1970s with the widespread use of MOS integrated circuit chips, enabling the production of advanced semiconductor devices, communication components, and sophisticated computer processors [1]. ICs are categorized based on the quantity of transistors or gates they incorporate: Small-Scale Integration (SSI) typically uses a few tens of transistors, Medium-Scale Integration (MSI) involves hundreds, and Large-Scale Integration (LSI) integrates thousands of transistors. However, VLSI surpasses LSI, MSI, and SSI by pushing integration to even greater levels, resulting in higher complexity. VLSI technology predominantly employs CMOS due to its exceptional noise immunity and minimal static power consumption [2, 3].

Before the era of VLSI technology, integrated circuits had limited capabilities, only able to execute a restricted set of functions. RAM, ROM, CPUs, and other logic components were built as separate electrical circuits. VLSI has revolutionized IC design by enabling the integration of all these functionalities into a single chip [4, 5]. VLSI relies on the system-on-chip architecture for IC development. VLSI circuits find applications in a wide range of devices, including personal computer and workstation microprocessors, graphics cards, digital cameras, camcorders, cell phones, portable computing devices, and embedded processors in automobiles. VLSI technology offers several benefits, including reduced circuit size, enhanced operating speed, reduced power consumption, and decreased device costs [6, 7].

The basic structure of FinFET is formed on a thin silicon substrate and gates are formed at vertical sides and fin act as channel connected between source and drain [8]. In FinFET technology there is a transition from planar to three-dimensional transistor design. Unlike planar MOSFETs with a flat channel, FinFET incorporate a fin-like structure that protrudes vertically from the substrate. This fin structure is surrounded by gate electrodes on three sides, allowing for enhanced control over the flow of current [9].

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II. EXISTING METHOD

A. 4 x 4 AM using FinFET transistors in GDI logic with 4-bit RCA

The 4x4 AM utilizing RCA is a sophisticated digital circuit designed for multiplying two 4-bit binary numbers [17, 18]. Within this circuitry, there are four distinct stages of 4-bit RCAs. Each stage processes the multiplication of specific input bits with respective binary digits. These interim multiplication results from each stage are subsequently accumulated, giving rise to the comprehensive multiplied outcome [15, 16]. Importantly, the outcome of one RCA stage serves a dual purpose: providing part of the multiplication result and acting as the carry-in for the subsequent RCA stage [19]. This design ensures that any carry-out generated during a multiplication is efficiently propagated through subsequent stages and the final product is realized through the accumulation of outputs and the last carry-out signal that is cascaded through the entire circuit.

B. 4x4 AM using FinFET transistors in GDI logic with 4-bit CLA

The implementation of a 4 x 4 AM leveraging CLA revolutionizes the efficiency of binary number multiplication. This innovative approach incorporates a grid-like structure where two 4-bit binary numbers are processed, significantly outperforming RCA by capitalizing on parallel carry computation. In this architecture, each cell within the array simultaneously computes partial products by multiplying the corresponding bits of the input numbers [20].

C. Wallace Tree Multiplier using FinFET transistors

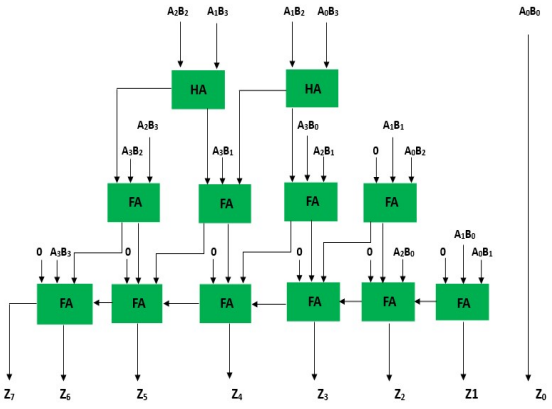


Fig.1. Schematic of **4 x 4** WTM

A Wallace Tree Multiplier (WTM) is a hardware multiplier architecture designed to enhance the speed of multiplication by reducing the number of partial products and utilizing a tree structure for efficient addition. The architecture of 4x4 Wallace Tree Multiplier is shown in Fig.1.The WTM operates by breaking down the multiplication process into stages, thereby reducing the number of partial products generated. In a 4x4 Wallace Tree Multiplier, for multiplying two 4-bit binary numbers, the multiplication process begins with the generation of partial products through bitwise multiplication of each bit of one operand with all bits of the other operand [22]. The next stage involves grouping these partial products into three sets based on the number of '1' bits in each group [24]. The partial products are then added in a parallel fashion within each group, followed by a reduction stage where the results from the previous stage are further added in a tree-like structure [24, 25].

III. PROPOSED METHOD

This section deals with the design of 4 x 4 AM using CLA in PTL Logic. The advantage of CLA over RCA lies in its ability to execute high-speed, efficient multiplication operations, reducing time complexity and overall computational delays. The architecture's efficacy in speeding up the multiplication process makes it an invaluable asset in high-performance computing scenarios, offering accelerated and efficient multiplication suitable for various applications in digital systems, including processors, digital signal processors, and arithmetic units. This approach stands as a testament to the advancements in computational efficiency and speed, presenting a pivotal shift from traditional RCA-based designs to more sophisticated parallel computation techniques. The architecture of 4x4 AM using CLA is displayed in Fig. 2.

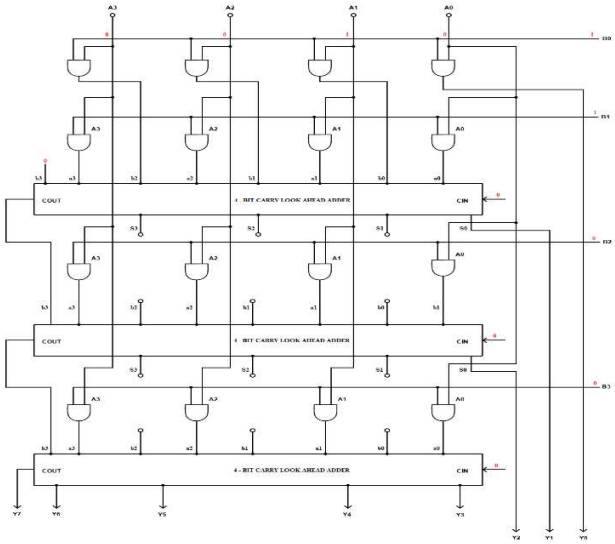


Fig. 2 Architecture of 4x4 AM using CLA

The schematic of 2 input AND, OR and EX-OR gates using PTL logic are displayed in Fig. 3 and Fig. 4.

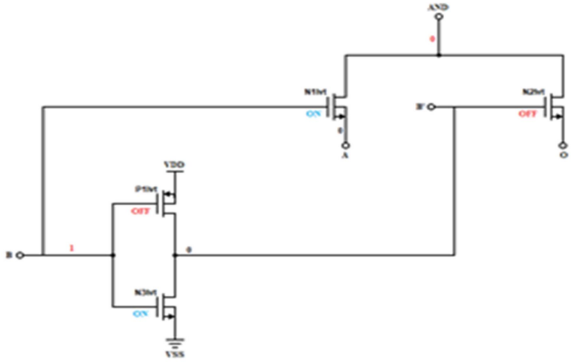


Fig. 3 Schematic view of AND gate using PTL logic

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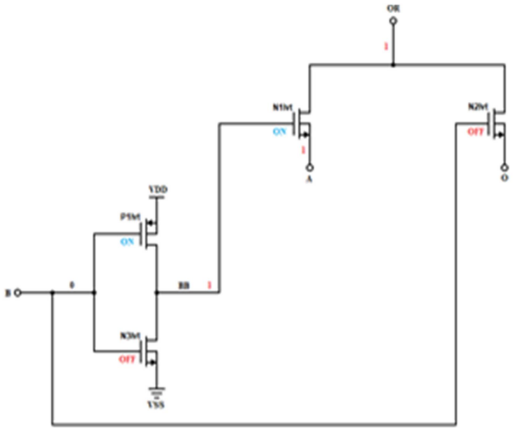


Fig. 4 Schematic view of OR gate using PTL logic

IV. Result And Discussion

A. *Simulation results of 4x4 AM using CLA*

The input (A0-A4 and B0-B4) and output waveforms (Y0-Y7) corresponding to 4x4 AM using CLA with PTL logic and with 18nm FinFET transistors are displayed in Fig. 5 and Fig. 6 respectively.

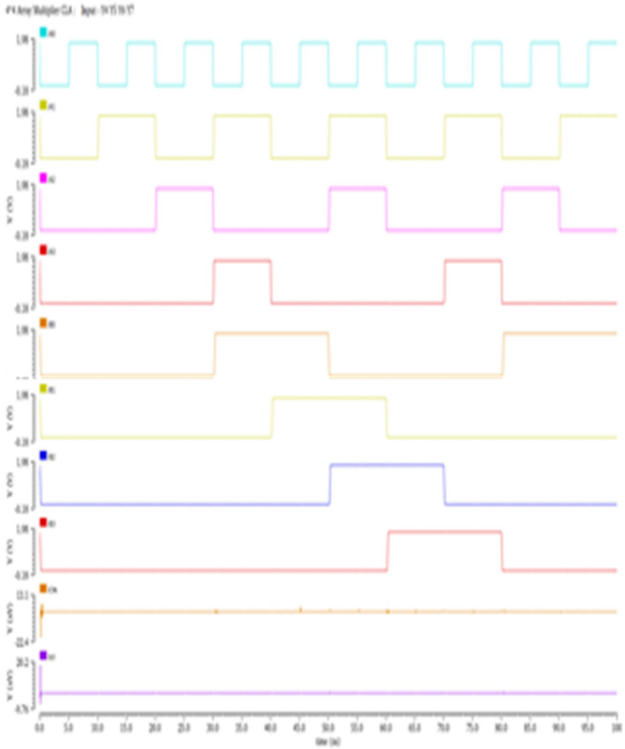


Fig. 5. Input waveforms of proposed 4 x 4 AM

B. *Layouts*

The layout of 2-input AND gate designed using Cadence Virtuoso Layout XL tool is shown in Fig. 7. It consists of two NMO transistors (for pulling down) and a PMOS transistor (for pulling up). Metal wire has a thin blue line, whereas polysilicon wire is green. Polysilicon and metal oxide are not directly connected to each other.

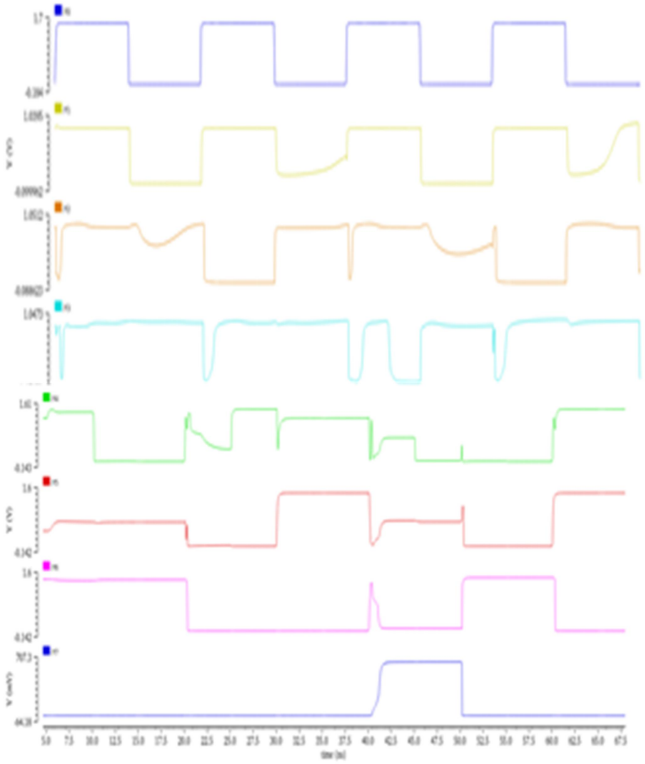


Fig.6. Output waveforms of proposed 4 x 4 AM

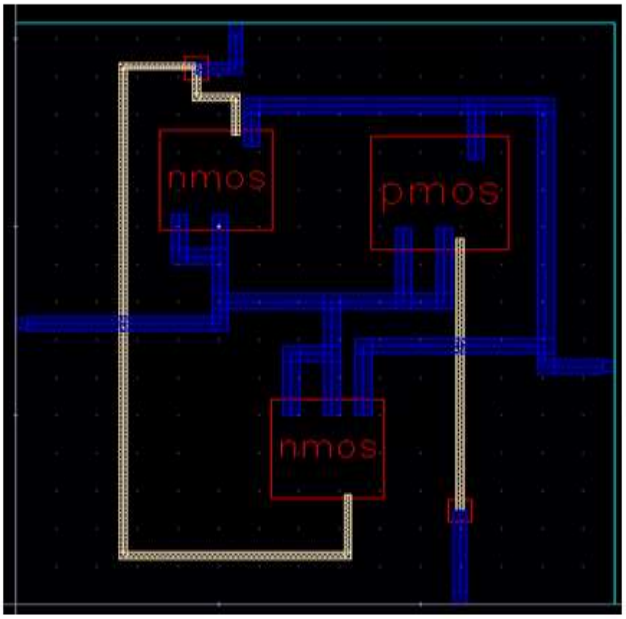


Fig 7 Layout of 2 input AND gate

V. Parameter Analysis

Parameters including Power, Propagation Delay, and Area in terms of Transistors count and Power Delay Product for 4 x 4 AM, WTM using 4 – bit RCA and CLA during operation in 18nm FinFET technology are tabulated in Table 1 and the graphical representation are shown in Fig. 8 and Fig.9.

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Table 1. Parameter analysis of multipliers

Multiplier (45nm)	Area (Transistor Count)	Power (µW)	Delay (ns)	PDP (pJ)
AM using RCA with GDI	152	724.43	30.26	21921.25
AM using CLA with GDI	440	886.319	10.25	9085.73
WTM using GDI	356	188.08	20.36	3829.30
WTM using PTL	324	448.66	10.16	4558.38
Proposed AM using CLA - PTL	832	185.073	10.10	1869.20

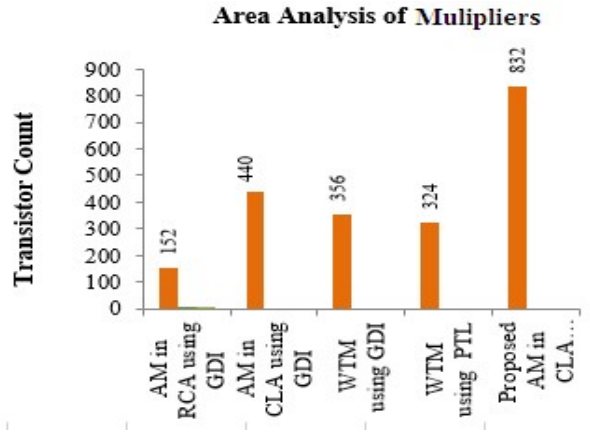


Fig. 8 Area analysis of multipliers

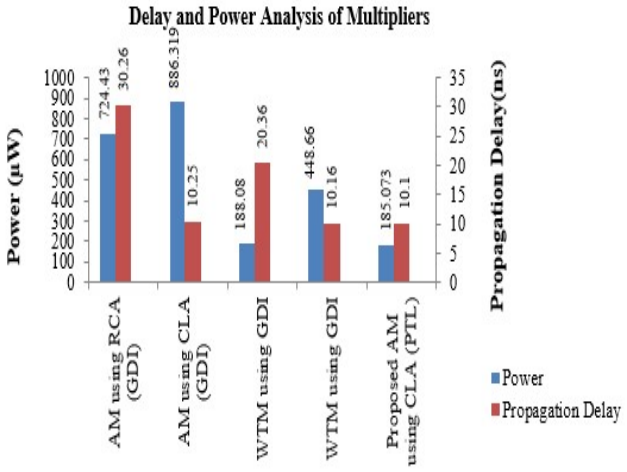


Fig. 9. Delay and power analysis of multipliers

The percentage improvement in speed and power of proposed multiplier when compared to other multiplier circuits are presented in Table 2 and the corresponding chart is shown in Fig.10.

Table 2. % Speed and power improvement of proposed AM

Proposed multiplier when compared to	% Delay improvement	% Power improvement
AM using RCA with GDI	66.62	74.45
AM using CLA with GDI	1.46	79.12
WTM using GDI	50.39	1.60
WTM using PTL	0.59	58.75

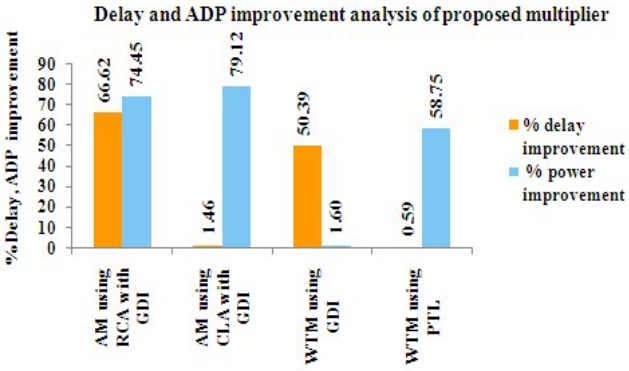


Fig.10. % Speed and Power improvement of proposed AM

A. Monte Carlo Analysis

Monte Carlo simulation is utilized for analysing and assessing the impact of variations or uncertainties in the manufacturing process of integrated circuits such as transistor dimensions, doping profiles, material properties, etc. The Monte Carlo histogram of AM using RCA and CLA for various samples is shown in Fig. 11 and Fig. 12.

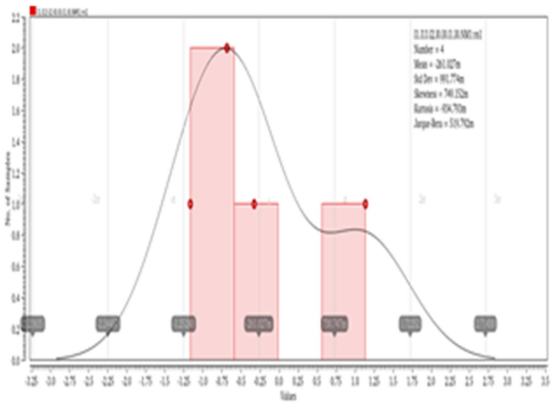


Fig.11. Monte Carlo histogram of existing AM using RCA

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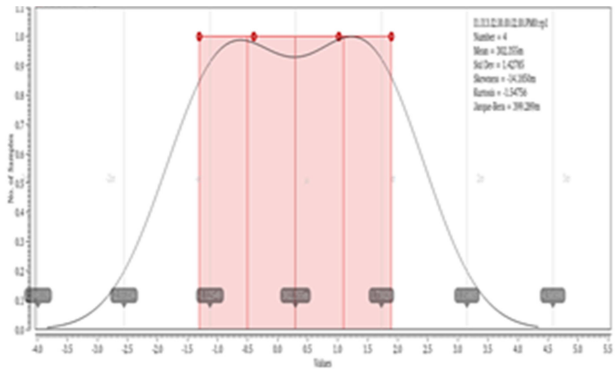


Fig.12. Monte Carlo histogram of proposed AM using CLA

VI. CONCLUSION

AM are used in digital signal processing tasks like digital filtering, image processing, and video processing. In the comparative analysis of 4 x 4 AM using CLA and RCA in the context of different logic such as PTL and GDI technique several key metrics were evaluated. The implementation of AM using CLA with PTL logic demonstrated advantages with lower propagation delay of 10.10 ns and power consumption of 185.07 µW. This shows that the proposed multiplier exhibits less delay and lower power consumption compared to other FinFET based multipliers. In addition, the proposed multiplier yields 66.62%, 1.46%, 50.39%,0.59% improvement in speed and 79.12%, 79.12%, 1.60%, 58.75% improvement in power when compared to AM using RCA with GDI technique, AM using CLA with GDI technique, WTM using GDI technique and WTM using PTL logic. This highlights the potential of AM using CLA with PTL logic for improved efficiency in signal and image processing applications.

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